Lab Assignment 02

Intermediate Biomedical Data Science (EN.580.375)

Overview

This lab practices statistical inference and multiple testing from Lectures 4–5. You will practice sampling variability, calculating confidence intervals, performing hypothesis tests, fitting linear models, and correcting for multiple hypothesis tests.

Groups are allowed, but each student must submit their own code and short write-up. If you work with others, identify your collaborators and write your responses in your own words. Include your code, plots, numerical results, and short explanations in the submitted notebook or equivalent file.

You may use Google Colab or a local Jupyter environment. The code chunks below are starter code: run and adapt them as needed, and include your code, plots, numerical results, and short explanations in the submitted notebook or equivalent file.

Suggested schedule

- **Wednesday:** Part 1 — sampling variability, confidence intervals, hypothesis tests, and linear models
- **Friday:** Part 2 — Type I and Type II errors, power, multiple testing, and genomic data visualizations

Learning goals

By the end of the lab, you should be able to:

- simulate sampling distributions and apply the Central Limit Theorem
- calculate and interpret standard errors, confidence intervals, and two-sample Welch's t-tests
- connect two-sample tests to linear models and Gaussian maximum likelihood
- distinguish Type I and Type II errors and estimate power
- apply Bonferroni and Benjamini–Hochberg corrections and choose between FWER and FDR
- interpret p-value histograms and volcano plots, and communicate conclusions with appropriate uncertainty and limitations

Part 1 — Wednesday: Inference from samples

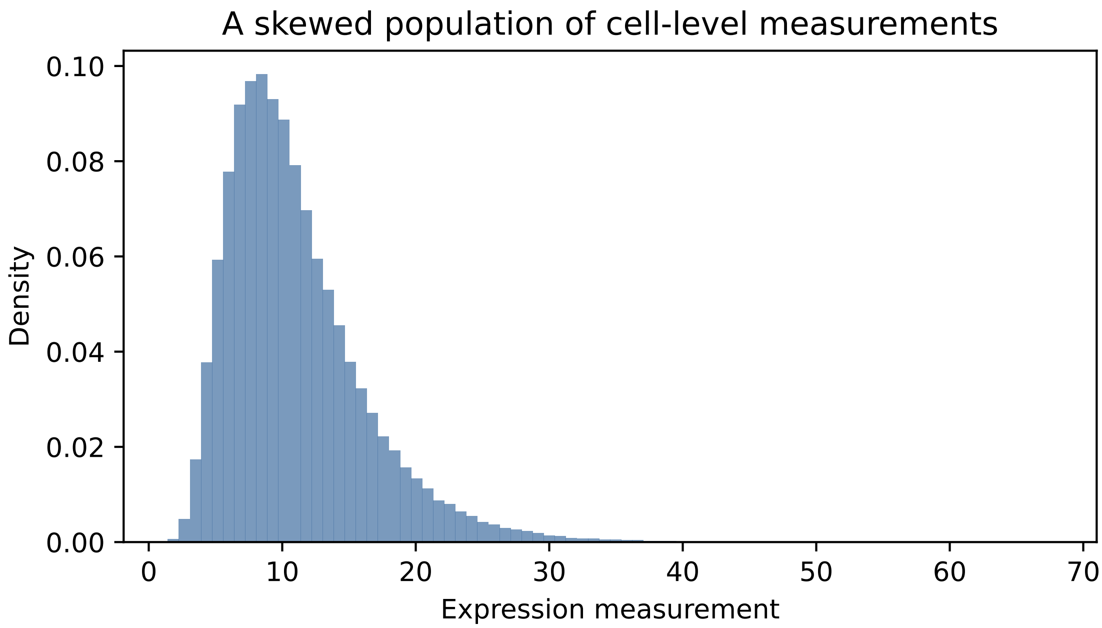
1. Sampling variability and confidence intervals

Start with a population of simulated cell-level expression measurements. The population is intentionally right-skewed, so the individual observations are not Gaussian.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
```

```python
rng = np.random.default_rng(375)
population = rng.lognormal(mean=2.3, sigma=0.45, size=100_000)

fig, ax = plt.subplots(figsize=(6, 3.5))
ax.hist(population, bins=80, density=True, color="#4E79A7", alpha=0.75)
ax.set_xlabel("Expression measurement")
ax.set_ylabel("Density")
ax.set_title("A skewed population of cell-level measurements")
plt.tight_layout()
```



Questions

1. Describe the shape of the population distribution. Does it look Gaussian?
2. For each n in `sample_sizes = [5, 20, 100]`, draw 5,000 independent samples from `population`, store their means, and plot the three sampling distributions. How do their centers and spreads change as n increases?
3. Explain how the simulation illustrates the Central Limit Theorem. Compare the simulated standard deviations with $s/\sqrt{n}$, and explain how standard errors change when sample size is multiplied by four.
4. Does the Central Limit Theorem claim that individual measurements become Gaussian as n increases? Explain the distinction.

Use the following seven control-cell measurements to examine uncertainty in a mean:

$$y_C = (8.8, 9.6, 10.2, 10.7, 11.0, 11.4, 12.1).$$

5. Calculate the sample mean, sample standard deviation, estimated standard error, and an approximate 95% confidence interval using

$$\bar{y} \pm t_{0.975,n-1} \frac{s}{\sqrt{n}}.$$

6. In your own words, explain what 95% confidence means for the procedure that produced this interval. What would be an incorrect interpretation?

7. Repeat the interval calculation using a much larger simulated sample from `population`. Why is the larger-sample interval generally narrower? Name two other factors that affect interval width, and distinguish uncertainty in the mean from variability among observations.

2. Comparing two groups

Treat the following as expression measurements from two independent groups:

```
control = np.array([9.2, 9.8, 10.0, 10.4, 10.7, 11.1, 11.5])
treated = np.array([10.1, 10.7, 11.0, 11.4, 11.6, 12.3, 12.5])
```

Use a plot and Welch's two-sample t-test to compare the groups.

Questions

1. Plot the individual observations and group means. Why is it useful to show the individual observations rather than only the means?
2. Calculate the estimated difference in means, its standard error, a two-sided test statistic, a p-value, and a 95% confidence interval.
3. State the null and two-sided alternative hypotheses in terms of the population means.
4. Interpret the p-value as a probability under the null model. What does it not mean?
5. Explain why a confidence interval for the difference can be more informative than a binary "significant/not significant" label.
6. Suppose another study estimates a larger difference in means but has a much larger standard error. Could its p-value be less significant than the result here? Explain using the signal-to-noise structure of the t statistic.
7. Why is it important to decide whether a test should be one-sided or two-sided before examining the result?

3. Linear models and likelihood

Questions

1. Fit the model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad x_i = 0 \text{ for control and } x_i = 1 \text{ for treated.}$$

   What do β_0 and β_1 represent? Explain why testing $H_0 : \beta_1 = 0$ asks the same scientific question as testing equal group means.

2. In one or two sentences, explain why maximizing a Gaussian likelihood with respect to β is equivalent to minimizing

$$\|y - X\beta\|_2^2$$

3. Give two assumptions that make a two-sample t-test or Gaussian linear model more reliable. For each assumption, describe one possible consequence of violating it.
4. Why should a plot of the observations be part of the analysis before reporting a test statistic or p-value?
5. If the group indicator were coded as $x_i = 1$ for control and $x_i = 0$ for treated instead, how would the interpretations of β_0 and β_1 change?
6. What does a residual represent in a linear model? Give one example of a residual pattern that might suggest the model is inadequate.

Part 2 — Friday: Multiple testing and genomic associations

4. Type I and Type II errors and power

Questions

1. For the two-group test in Problem 2, describe a Type I error and a Type II error in the context of treatment and expression.
2. Define statistical power as a probability. What event is the test trying to detect?
3. Use simulation to estimate the power of a two-sample test as the treated-group mean changes while the control-group mean and sample sizes stay fixed. Plot power against the true treatment effect for at least five effect sizes.
4. Repeat the power simulation after doubling both group sample sizes. What changes, and why?
5. Explain why lowering the significance level from $\alpha = 0.05$ to $\alpha = 0.01$ generally decreases Type I errors but can increase Type II errors.
6. Explain why a larger effect, larger sample size, or lower measurement noise generally increases power.

5. FWER and Bonferroni correction

Use this small set of p-values:

```
p_values = np.array([0.001, 0.008, 0.012, 0.021, 0.047, 0.18, 0.44, 0.72])
```

Questions

1. At family-wise level $\alpha = 0.05$, which hypotheses would be rejected without correction? Which would be rejected with the Bonferroni threshold α/m?
2. Explain what FWER controls and why Bonferroni is called conservative.
3. Use the Bonferroni-adjusted values $\min(mp_j, 1)$ to produce adjusted p-values. Confirm that thresholding these at 0.05 gives the same decisions as thresholding raw p-values at $0.05/m$.
4. Give one setting in which controlling FWER might be preferable to controlling FDR.

6. False discovery rate and Benjamini–Hochberg

Apply the Benjamini–Hochberg procedure at target $q = 0.05$ to `p_values`.

Questions

1. Sort the p-values and calculate the thresholds kq/m for each rank k.
2. Find the largest rank satisfying $p_{(k)} \leq kq/m$. Which hypotheses are discoveries?
3. Compare the Bonferroni and BH discovery lists. Explain why they can differ.

4. Define the false discovery proportion (FDP) for one dataset and the false discovery rate (FDR) across repeated datasets. Why should we not interpret FDR as a guarantee that exactly 5% of this particular list are false?
5. Write a short recommendation: would FWER or FDR be more appropriate for an exploratory screen of thousands of genes, and why?

7. P-value histograms and genomic plots

Simulate a gene-level results table containing 2,000 genes. Include a mostly-null group of p-values and a smaller group of alternatives with p-values concentrated near zero. Also simulate a log fold change for each gene.

```
rng = np.random.default_rng(375)
n_genes = 2000
n_alt = 250
p_null = rng.uniform(0, 1, n_genes - n_alt)
p_alt = rng.beta(0.35, 1, n_alt)
p_all = np.r_[p_null, p_alt]
log2_fc = rng.normal(0, 0.35, n_genes)
log2_fc[-n_alt:] += rng.choice([-1, 1], size=n_alt) * rng.uniform(0.8, 2.0, n_alt)
```

Questions

1. Plot a histogram of all p-values. What would a flat background represent? What does an excess near zero suggest?
2. Create a volcano plot with $\log_2$ fold change on the horizontal axis and $-\log_{10}(p)$ on the vertical axis. Add guide lines at $|\log_2 \mathrm{FC}| = 1$ and $p = 0.05$. Which points have both large effects and small raw p-values?
3. Explain why points passing the raw $p < 0.05$ line in a volcano plot are not automatically valid discoveries when thousands of genes are tested.
4. List the minimum fields you would report for each gene in a defensible analysis. Include an effect estimate, raw p-value, and multiple-testing-adjusted p-value.

Final synthesis

8. From data to a scientific conclusion

Choose one result from this lab and write a short paragraph addressing:

1. What scientific question were you asking?
2. What quantity did you estimate or test?
3. What did the confidence interval, p-value, or multiple-testing adjustment contribute to your conclusion?
4. What uncertainty, assumption, or limitation should accompany the conclusion?
5. What additional data or analysis would make the conclusion more reliable?

9. Reflection

Answer briefly:

1. Which concept from Wednesday was most intuitive, and why?
2. Which concept from Friday was most intuitive, and why?

3. Which concept remains unclear?
4. Describe one connection between sampling variability, hypothesis testing, and multiple testing that became clearer through the lab.

Submission checklist

- a notebook or equivalent file containing your code;
- responses to all questions in both the Wednesday and Friday parts;
- labeled and interpretable figures;
- numerical results with appropriate interpretation;
- a results table summarizing selected estimates, p-values, and adjusted p-values;
- a final synthesis paragraph;
- a brief reflection; and
- a note identifying collaborators, if you worked in a group.